

Patterns & Machine Learning

Discovering how machines learn just like us!



What Are Patterns?

Patterns are things that happen again and again!

Look around you — patterns are everywhere!
In your clothes, in rangoli designs, in music, and even in nature.



Patterns in Our Clothes

Look at the beautiful patterns in our clothes!
Checks, stripes, floral prints — Indian fabrics are full of amazing patterns!



Patterns in Shapes & Colors

Shapes and colours can make patterns too!

Red, blue, red, blue... Circle, square, circle, square...

Can you see the pattern in beads on a bracelet?



Patterns in Sounds & Songs

We hear patterns in music and clapping!

Clap, clap, pause, clap, clap, pause!

Songs have patterns too – the chorus repeats again and again.



Finding Patterns in Our Day

Our daily routine is a pattern!

Wake up, brush teeth, eat breakfast, go to school, come back, play, sleep.

What part of your day happens every day?



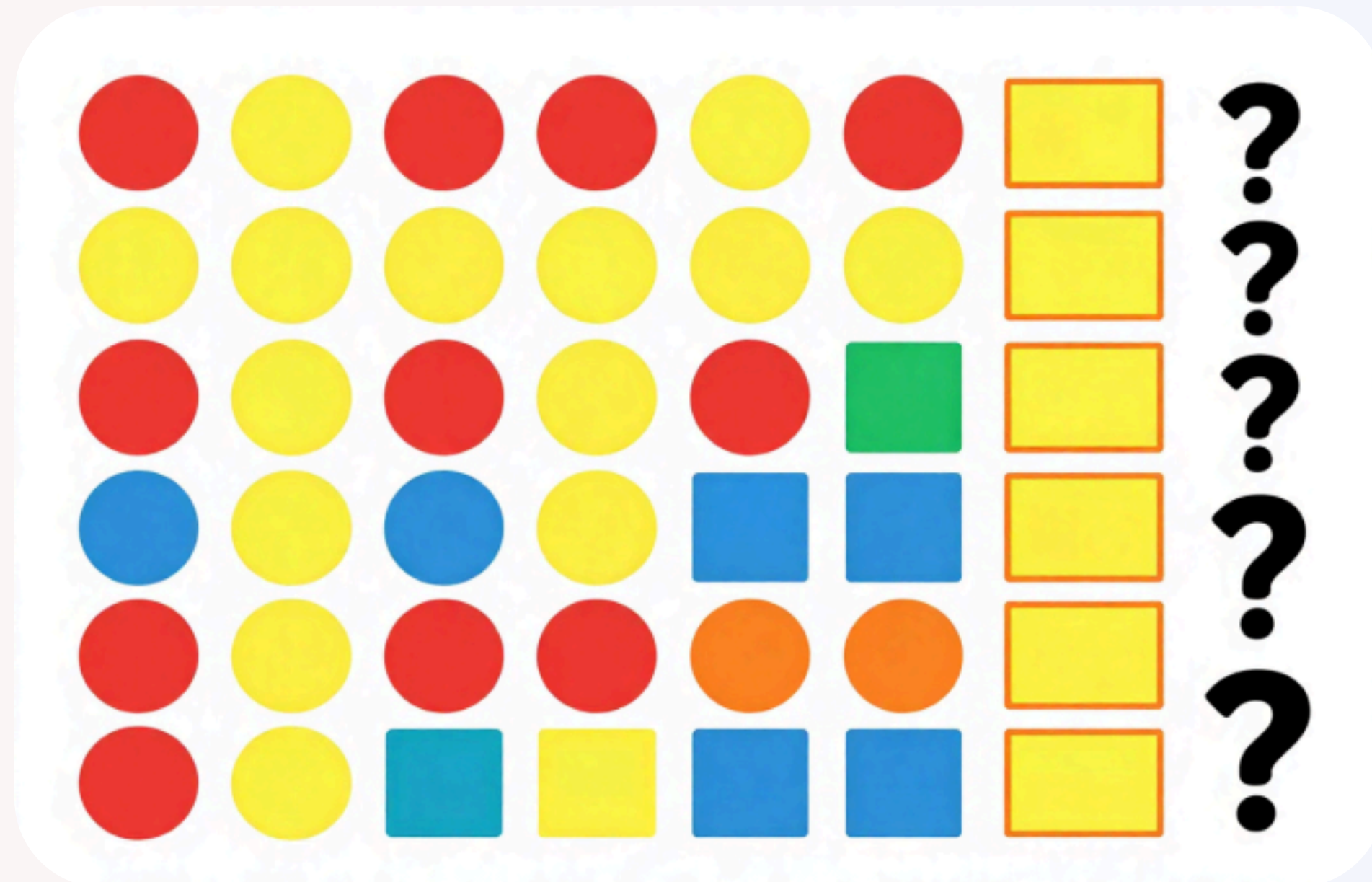
Let's Play: What Comes Next?

Can you guess the missing piece?

Red - Yellow - Red - Yellow - ?

Circle - Square - Circle - Square - ?

Think and guess what comes next!



Machines and Patterns

Machines can learn by finding patterns – just like you!

A computer looks at many pictures and finds what is the same.
Example: It learns that cats have pointy ears and whiskers!

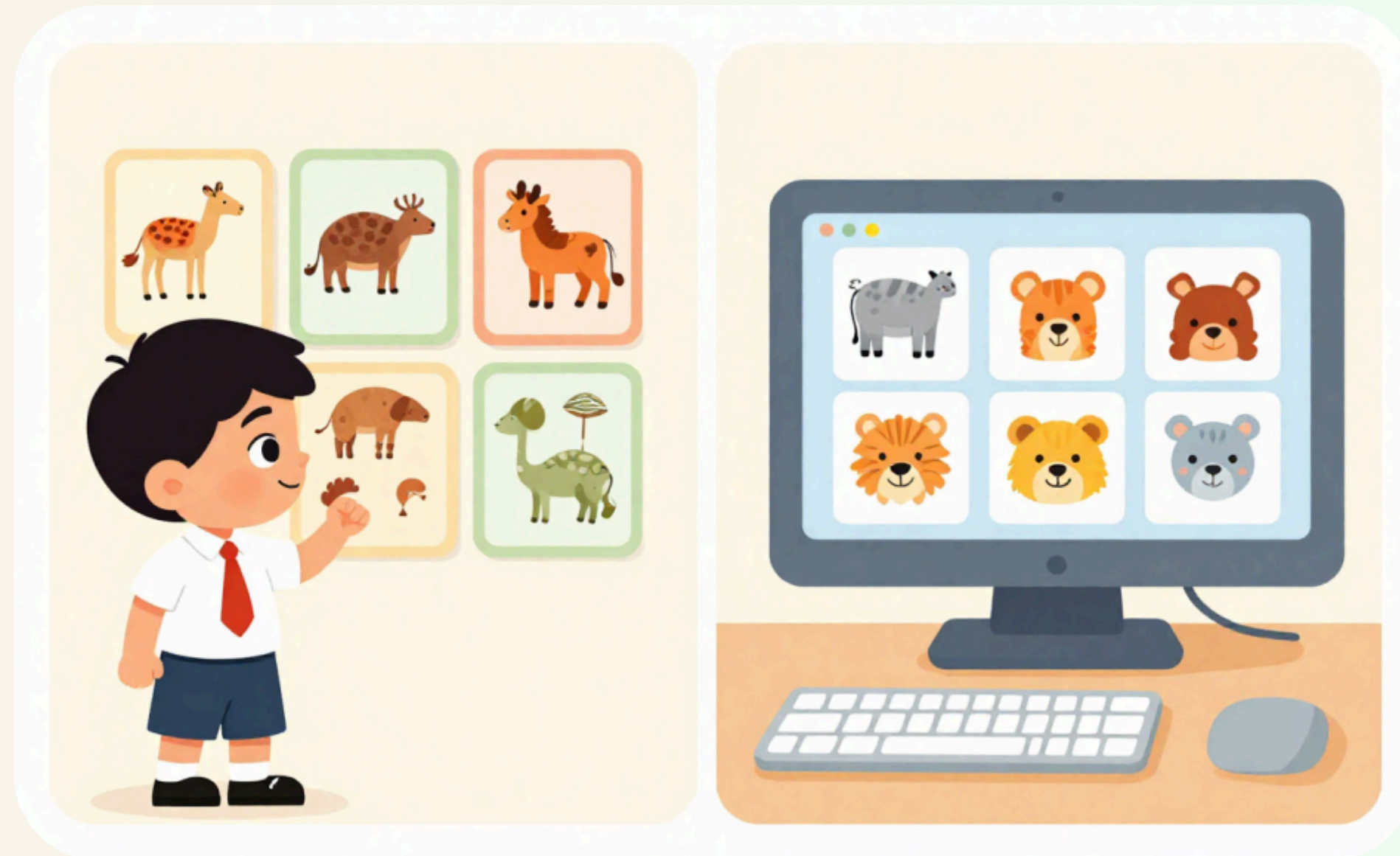


How Do Machines Learn?

Humans and machines both learn from patterns!

You learn by seeing and practising.

A machine learns by looking at many, many examples and pictures!



Machine Learning in Daily Life

Machines that use patterns are all around us!

Voice assistants understand your speech.

Phone cameras recognise faces.

Games get smarter as you play!



Patterns Help Us Solve Problems

When we see a pattern, we can guess what happens next!

If it rains every day at 5 PM, what should we do?
Take an umbrella! That is using a pattern to solve a problem.



Let's Make Our Own Pattern!

Now it is your turn to create a pattern!

Use coloured blocks, beads, or crayons.
Make a pattern and show it to your friend!



Activity: Spot the Pattern!

Look at the sequences below and guess what comes next!

Fruit colours: Apple, Banana, Apple, Banana, ?

Animal patterns: Spots, Stripes, Spots, Stripes, ?

Rangoli: Red, Yellow, Red, Yellow, ?

